Composite Structures 300 (2022) 116131

Contents lists available at ScienceDirect

Composite Structures

journal homepage: www.elsevier.com/locate/compstruct

Building block design for composite metamaterial with an ultra-low thermal expansion and high-level specific modulus

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ARTICLE INFO

Keywords:
Composite metamaterials
The building block assembly method
The unprecedented thermal–mechanical stability
High-level lightweight
The remarkable specific modulus

ABSTRACT

The metamaterials with unique thermal expansions are attracting increasing interest in a broad range of applications. Despite numerous metamaterials with unique thermal expansions have been demonstrated, none of them is without inherent laminations, either in terms of their limited design freedom, the immutability feature of the specimen once fabricated, and the low specific modulus. Inspired by the building block (e.g., Lego toys), this paper introduces a generalized design strategy for reconfigurable mechanical metamaterials with unique thermal/mechanical performances. The triangle metamaterial constructed by the specific collection of structural building blocks shows the capability of this design for achieving an unprecedented low effective CTE (i.e., 0.07 ppm/°C), high-level lightweight (i.e., 0.549 g/cm^3), and a remarkable relative specific modulus (i.e., 41385.55 kN•mm/kg). More specifically, the assembly method with well-designed connectors presented here promises the mechanical robustness and reconfigurable feature of metamaterials. Additionally, the thermal–mechanical factor Φ is adopted for accurately characterizing compatibility of thermal deformation and mechanical properties of the metamaterial. Finally, the comparison of the absolute value of CTE and the comprehensive thermal–mechanical property serves a quantitative comparison of the mechanical metamaterials demonstrated here to the previous reported results, indicating the capability of our designs for their practical engineering applications.

1. Introduction

The mechanical metamaterials with tunable thermal expansion behaviors are attracting increasing interest to their extensive applications in the fields of requiring stability in geometrics to maintain their shapes upon a temperature change, such as MEMS [1–6], flexible electronics [4,5,7–12], optical components [13,14], the antenna platform of satellites [15–17], and even biomedical components (e.g., dental fillings) [11,18,19]. To address these demands, numerous efforts have been devoted, and diverse metamaterial designs have been proposed. Unlike the natural materials with constant thermal expansions and specific CTE values (-9.3 $10.3\,ppm/^{\circ}$C), these metamaterials are mainly constructed by two engineering materials with different CTEs, and allow access to a tunable thermal expansion performance by tailoring their geometric configurations [15,20–27]. The underlying mechanisms arise from the thermal strain mismatch between different constitute materials upon a temperature increase. We can categorize the previously demonstrated metamaterials into five major categories in terms of thermal-induced deformation mode: the stretching-dominant design [15,28–35], the bending-dominant design [36–43], the stretch-bending coupling dominate design [2,3,22,24], the design based on the topological method [44,45], and the composite metamaterial filled by the nanoparticles with negative CTE [21,46–48]. Despite numerous metamaterials with unique thermal expansion (i.e., positive or negative CTE in a giant tunable range, or nearly zero CTE) have been demonstrated in terms of theoretical predictions, Finite Element Analysis (FEA) results and experiments, their practical engineering application is still a challenge, considering their certain deficiencies in the aspects of the limited design freedom, the insufficient mechanical robustness at the connections between different consistent materials, the immutability feature of the specimen once fabricated, and the low specific modulus. However, it is still challenging to achieve a mechanical metamaterial with combined contributes of the ultra-low CTE (e.g., <0.1 ppm/°C), high-level lightweight (e.g., <1.0 g/cm^3), and high-level specific modulus (e.g., >1 ×

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https://doi.org/10.1016/j.compstruct.2022.116131
Received 15 January 2022; Received in revised form 29 July 2022; Accepted 13 August 2022
Available online 19 August 2022

$10^4\ kN \bullet mm/kg$). Additionally, enhanced robustness of connections between two structural building blocks always represent a crucial factor in the designing of these metamaterials, especially when the mechanical metamaterials are exposed to a cyclic thermal heating load. In the previously reported researches, connection methods through dovetail connectors and super glue with strong adhesion were adopted to form mechanical metamaterials. However, the inherent deficiencies arising from the aforementioned connection approaches, such as immutability of the structures after one-time molding or once prepared [22,37–39,49,50] and the narrow accessible temperature window [34,38,50,51], set practical constraints on the applications of these metamaterials in the fields that require low cost, recyclable feature, high load-bearing capacity and thermal–mechanical stability upon a temperature change.

To address these demands, the design strategy inspired by the building block assembly method (e.g., Lego toys) is presented here to achieve the reconfigurable mechanical metamaterials with ultra-low CTE, lightweight and high-level specific modulus. The triangle metamaterial serving as the representative demonstrations are systemically studied in the terms of theoretical predictions, FEA results, and experiments. These metamaterials are constructed by the specific collection of structural building blocks (i.e., solid and hourglass lattice-filled beams in frame and actuation material, and well-designed connectors in actuation material). Combined with theoretical predictions, FEA, and experiments, these designs show the capabilities of this design in achieving an unprecedented low absolute value of effective CTEs (i.e., 0.07 ppm/°C), high-level lightweight (i.e., 0.549 g/cm^3), high-level specific modulus (i.e., 41385.55 kN•mm/kg), and reconfigurability. Notably, the unique assemble method with well-designed connectors and bolts presented here promises the mechanical robustness of metamaterials upon the cyclic thermal heating load and enables the reconfigurable feature of the fabricated metamaterials to address the other potential demands. Additionally, the thermal–mechanical factor Φ is adopted for accurately characterizing compatibility of thermal and mechanical properties of the metamaterial. Finally, the comparison of the CTE serves a quantitative comparison of the mechanical metamaterials demonstrated here to the previous reported results, indicating the capability of our designs for their practical engineering applications.

2. Results and discussion

2.1. Design strategies of the triangle metamaterial inspired by the building block assembly method

Fig. 1 (a) schematically illustrates the design strategy of the triangle metamaterial with a tunable thermal expansion and a remarkable thermal–mechanical stability. The metamaterial is arranged in a 2×2 periodic array of unit cells, each of which is composed of three solid beams in frame material (in yellow with a lower CTE) and actuation material (in red with a higher CTE), and three well-designed connectors in actuation material. Here, specific connectors patterned in a triangle shape are adopted to join the neighboring beams through bolt joints, promising the mechanical robustness of metamaterials. As shown in Fig. 1(b)-(c), the mechanical and thermal expansion performances of the metamaterial can be characterized by the following geometric parameters: $L_{Actuation}$, $W_{Actuation}$, $T_{Actuation}$, $l_{Actuation}$, $w_{Actuation}$, $t_{Actuation}$, L_{Frame}, W_{Frame}, T_{Frame}, l_{Frame}, w_{Frame}, t_{Frame}, and the angle of the metamaterial β. Here, $L_{Actuation}$ and L_{Frame} are the lengths of actuation and frame beams, while $W_{Actuation}$, W_{Frame} and $T_{Actuation}$, T_{Frame} are their corresponding widths and thicknesses, respectively. Similarly, $l_{Actuation}$ and l_{Frame} are the lengths of regions of the beams joining with connectors, while $w_{Actuation}$, w_{Frame} and $t_{Actuation}$, t_{Frame} are their corresponding widths and thickness, respectively. For simplification, the relationships along these geometric parameters are fixed as: $W_{Actuation} = W_{Frame}$, $T_{Actuation} = T_{Frame} = t_{Actuation} = t_{Frame} = W_{Frame}$, and $w_{Actuation} = w_{Frame} = 2mm$, and $l_{Actuation} = l_{Frame} = 10mm$, respectively. Additionally, the lengths of the connector $l_{Connector}$ is fixed as $2l_{Actuation}$, while the thicknesses $t_{Connector}$ equals to the ones of actuation and frame beams, respectively. The length of the beam in frame material L_{Frame} is $100mm$ herein, and the length of actuation beam $L_{Actuation}$ can be geometrically given by $L_{Actuation} = \frac{(2L_{Frame}+4l_{Connector})}{\sin\beta} - 2l_{Connector}$. Additionally, considering the demands of lightweight and high specific modulus, the triangle metamaterial can be also further optimized by filling the solid beams with periodic hourglass lattices in the corresponding materials themselves (as shown in Fig. 1 (c)). The hourglass lattice structure as the fundamental element of the triangle metamaterial composed of lattice-filled beams be characterized by the length L_{Rod}, radius r and the angle θ of the rod in the hourglass

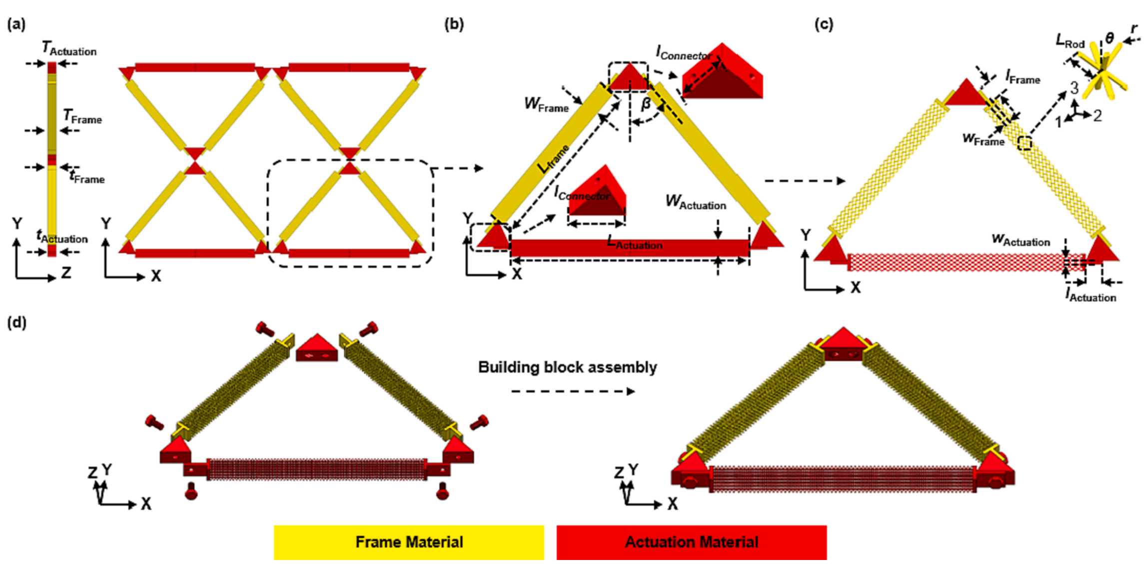

Fig. 1. Design strategies of the triangle metamaterial inspired by the building block assembly method. (a) Schematic illustration of the triangle metamaterial in a 2x2 array. (b) Illustration of one unit cell of the mechanical metamaterial consisting of solid beams and its key geometric parameters; (c) Illustration of one unit cell of the mechanical metamaterial constructed by hourglass lattice-filled beams and its key geometric parameters; (d) Schematic illustration of the design strategies inspired by building block assembly method for a triangle mechanical metamaterial constructed by the hourglass lattice-filled beams in frame material (in yellow) and actuation material (in red) before (left) and after (right) assembled through bolts. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

lattice, which are given as $2.5\sqrt{2}mm$, $0.25mm$, 30°, respectively. Fig. 1(d) illustrates the assembly strategy inspired by building block assembly method for a triangle mechanical metamaterial before (left) and after (right) assembled through several bolts. Here, the well-designed connectors and bolts allow access to the robustness connection of the neighboring hourglass lattice-filled beams and the reconfigurable feature of the whole metamaterial.

2.2. Theoretical predictions of the effective CTE of the triangle metamaterial

The theoretical model is developed to predict the mechanical and thermal deformation of the triangle metamaterial, and describes the dependences of effective CTE on the geometric parameters (as shown in Fig. 2(a)). The strain mismatch due to the difference in the mechanical and thermal properties of constituting materials causes the bending deflection of the metamaterial upon a mechanical loading or temperature change, resulting in the unique deformation response of the metamaterial. Triangle metamaterial is composed of three beams in two different materials (i.e., frame material in yellow and actuation material in red) and three well designed connectors in actuation material. Considering the geometric symmetry of the unit cell, only one half of horizontal cantilever beam in actuation material along the X- direction and an incline cantilever beam in frame material are analyzed. Thus, for the horizontal cantilever beam in actuation material (as shown in Fig. 2(b)), the deflection $\omega_{Actuation}^{Tri-T}$ and angle $\theta_{Actuation}^{Tri-T}$ caused by the concentrated force F_{Tri-T} and moment M_{Tri-T} due to thermal strain mismatch are expressed as.

$$\omega_{Actuation}^{Tri-T} = \frac{M_{Tri-T} \bullet \left(L_{Actuation}^{Tri-T}\right)^2}{8E_{Actuation} \bullet I_{Actuation}} \tag{1}$$

$$\theta_{Actuation}^{Tri-T} = \frac{M_{Tri-T} \bullet L_{Actuation}^{Tri-T}}{2E_{Actuation} \bullet I_{Actuation}} \tag{2}$$

Notably, $E_{Actuation}$ is $E_{Lattice-1}$ (or $E_{Lattice-2}$) for the hourglass lattice-filled beams in actuation material, or equals to the elastic modulus of actuation material for the solid beam. The effective modulus $E_{Lattice-1}$ and $E_{Lattice-2}$ of the lattice hourglass structure along the 1- and 2- directions have been predicted by measuring the mechanical deformation caused by a pair of compressive loads applied on the top and bottom surfaces in the previous work [49], and is $\frac{\sqrt{2}F}{\omega_X \bullet L_{Rod} \bullet \sqrt{1+\cos^2\theta \bullet \cos\theta}}$, where ω_X is the mechanical deflection of the rod in the lattice hourglass structure, and E is the modulus of the material in the lattice hourglass structure. $L_{Actuation}^{Tri-T}$ is the length of actuation beam after a temperature increase ΔT, and is expressed as.

$$L_{Actuation}^{Tri-T} = L_{Actuation} \bullet (1 + \alpha_{Actuation}\Delta T) \bullet \left(1 - \frac{F_{Tri-T}}{E_{Actuation} \bullet W_{Actuation} \bullet T_{Actuation}}\right) \tag{3}$$

For the incline cantilever beam in frame material (as shown in Fig. 2(c)), the deflection ω_{Frame}^{Tri-T} and angle θ_{Frame}^{Tri-T} caused by the concentrated force F_{Tri-T} and moment M_{Tri-T} due to thermal strain mismatch are expressed as.

$$\omega_{Frame}^{Tri-T} = \frac{F_{Tri-T-T} \bullet \left(L_{Frame}^{Tri-T}\right)^3}{3E_{Frame} \bullet I_{Frame}} - \frac{M_{Tri-T} \bullet \left(L_{Frame}^{Tri-T}\right)^2}{2E_{Frame} \bullet I_{Frame}} \tag{4}$$

$$\theta_{Frame}^{Tri-T} = \frac{F_{Tri-T-T} \bullet \left(L_{Frame}^{Tri-T}\right)^2}{2E_{Frame} \bullet I_{Frame}} - \frac{M_{Tri-T} \bullet L_{Frame}^{Tri-T}}{E_{Frame} \bullet I_{Frame}} \tag{5}$$

Here, $F_{Tri-T-N}$ and $F_{Tri-T-T}$ represent the projections of the concentrated force F_{Tri-T} along and vertical the axial direction of the cantilever beam. Notably, E_{Frame} equals to $E_{Lattice-1}$ (or $E_{Lattice-2}$) of the hourglass lattice-filled beam in frame material, or the elastic modulus E_F of frame material for the solid beam. L_{Frame}^{Tri-T} is the length of frame beam after a temperature increase, and is.

$$L_{Frame}^{Tri-T} = L_{Frame} \bullet (1 + \alpha_{Frame}\Delta T) \bullet \left(1 + \frac{F_{Tri-T-N}}{E_{Frame} \bullet W_{Frame} \bullet T_{Frame}}\right) \tag{6}$$

Note that due to the geometric constraint of the metamaterial after a temperature change ΔT, the relationship of the deformation and rotation angle in actuation beam and frame beam can be expressed as.

$$\begin{cases} L_{Actuation}^{Tri-T} - L_{Actuation} + 2l_{Connector} \bullet \alpha_{Actuation}\Delta T \\ = 2\omega_{Frame-X}^{Tri-T} + \dfrac{2L_{Frame} \bullet \alpha_{Frame}\Delta T + 4l_{Connector} \bullet \alpha_{Actuation}\Delta T}{\sin\left(\beta + \theta_{Frame}^{Tri-T}\right)} \\ \theta_{Actuation}^{Tri-T} = \theta_{Frame}^{Tri-T} \end{cases} \tag{7}$$

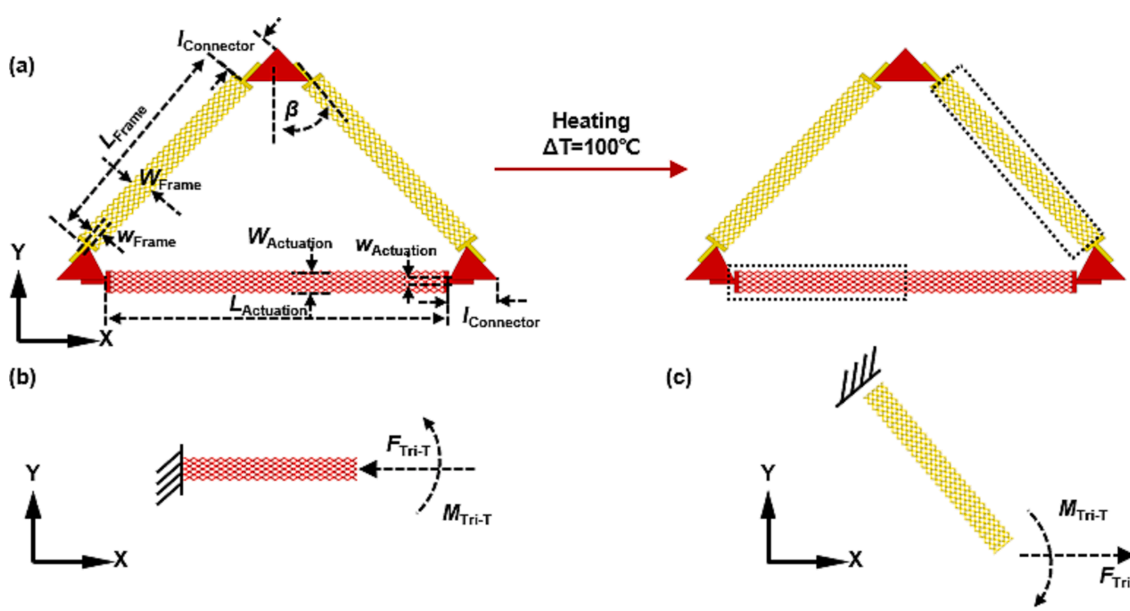

Fig. 2. Schematic illustration and theoretical analysis of the effective CTE of the triangle metamaterial. (a) FEA results of the unit cell before (left) and after (right) a temperature increase ($\Delta T = 100^{\circ}C$); (b-c) Mechanical boundary conditions of one half of horizontal beam in frame material and an incline beam in frame material after a temperature increase.

where $\omega^{Tri-T}_{Frame-X}$ is the projection of the deflection ω^{Tri-T}_{Frame} along the X-direction, and is $\omega^{Tri-T}_{Frame} \bullet \cos\left(\beta + \theta^{Tri-T}_{Frame}\right)$, while $\omega^{Tri-T}_{Frame-Y}$ is the projection along the Y- direction, and is $\omega^{Tri-T}_{Frame} \bullet \sin\left(\beta + \theta^{Tri-T}_{Frame}\right)$.

Thus, the relationship between the concentrated force F_{Tri-T} and moment M_{Tri-T} can be determined based on the above equations, and the dependence of deflections and rotation angles on the geometric parameters of metamaterials can be obtained. To summarize, the thermal deformation Δ_T of the metamaterial after the temperature change can be finally given as.

$$\Delta_T = \frac{L_{Frame} \bullet \alpha_{Frame}\Delta T + 2l_{Connector} \bullet \alpha_{Actuation}\Delta T}{cos\left(\beta + \theta^{Tri-T}_{Frame}\right)} + \omega^{Tri-T}_{Actuation} - \omega^{Tri-T}_{Frame-Y} \tag{8}$$

And effective thermal expansion coefficients of the triangle metamaterial $\alpha_{Triangle}$, can be expressed by.

$$\alpha_{Triangle} = \frac{\Delta_T}{h_{Triangle} \bullet \Delta T} \tag{9}$$

Here, $h_{Triangle}$ as the height of the triangle metamaterial along the Y-direction, is $\frac{L_{Frame}}{\cos\beta} + \frac{2l_{Connector}}{\cos\beta}$.

2.3. Theoretical predictions of the relative specific modulus of the triangle metamaterial

The effective modulus of the triangle metamaterial is predicted by analyzing the mechanical deformation caused by a concentrated force P on the metamaterial along the Y- direction (as shown in Fig. 3(a)). For simplification, one half of horizontal cantilever beam in actuation material and an incline cantilever beam in frame material are analyzed (as shown in Fig. 3(b) and (c)). For the cantilever beam in actuation material, the mechanical forces $P/2$, F_{Tri-F} and the moment M_{Tri-F} caused by the concentrated force P are subjected at the ends of beams, and the corresponding deflection $\omega^{Tri-F}_{Actuation}$ and angle $\theta^{Tri-F}_{Actuation}$ can be expressed as.

$$\omega^{Tri-F}_{Actuation} = \frac{P \bullet \left(L^{Tri-F}_{Actuation}\right)^3}{48E_{Actuation} \bullet I_{Actuation}} + \frac{M_{Tri-F} \bullet \left(L^{Tri-F}_{Actuation}\right)^2}{8E_{Actuation} \bullet I_{Actuation}} \tag{10}$$

$$\theta^{Tri-F}_{Actuation} = \frac{P \bullet \left(L^{Tri-F}_{Actuation}\right)^2}{16E_{Actuation} \bullet I_{Actuation}} + \frac{M_{Tri-F} \bullet L^{Tri-F}_{Actuation}}{2E_{Actuation} \bullet I_{Actuation}} \tag{11}$$

Here, $L^{Tri-F}_{Actuation}$ is the length of actuation beam stretched along the axial direction, and is.

$$L^{Tri-F}_{Actuation} = L_{Actuation} \bullet \left(1 + \frac{F_{Tri-F}}{E_{Actuation} \bullet W_{Actuation} \bullet T_{Actuation}}\right) \tag{12}$$

For the cantilever beam in frame material (as shown in Fig. 3(c)), the deflection ω^{Tri-F}_{Frame} and angle θ^{Tri-F}_{Frame} caused by the concentrated force $P/2$, F_{Tri-F} and moment M_{Tri-F} due to the mechanical loading are expressed as.

$$\omega^{Tri-F}_{Frame} = \frac{\left(P_{Tri-T} - F_{Tri-F-T}\right) \bullet \left(L^{Tri-F}_{Frame}\right)^3}{3E_{Frame} \bullet I_{Frame}} + \frac{M_{Tri-F} \bullet \left(L^{Tri-F}_{Frame}\right)^2}{2E_{Frame} \bullet I_{Frame}} \tag{13}$$

$$\theta^{Tri-F}_{Frame} = \frac{\left(P_{Tri-T} - F_{Tri-F-T}\right) \bullet \left(L^{Tri-F}_{Frame}\right)^2}{2E_{Frame} \bullet I_{Frame}} - \frac{M_{Tri-F} \bullet L^{Tri-F}_{Frame}}{E_{Frame} \bullet I_{Frame}} \tag{14}$$

Here, P_{Tri-N}, P_{Tri-T} and $F_{Tri-F-N}$, $F_{Tri-F-T}$ represent the projections of the concentrated force P and F_{Tri-F} along and vertical the axial direction of frame beam. L^{Tri-F}_{Frame} is the length of frame beam due to the mechanical loading, and is.

$$L^{Tri-F}_{Frame} = L_{Frame} \bullet \left(1 - \frac{F_{Tri-F-N} + P_{Tri-N}}{E_{Frame} \bullet W_{Frame} \bullet T_{Frame}}\right) \tag{15}$$

Note that due to the geometric constraint of the metamaterial, the deformation and rotation angle in actuation beam and frame beam are always consistent.

$$\begin{cases} L^{Tri-F}_{Actuation} - L_{Actuation} = 2\omega^{Tri-F}_{Frame-X} \\ \theta^{Tri-F}_{Actuation} = \theta^{Tri-F}_{Frame} \end{cases} \tag{16}$$

where $\omega^{Tri-F}_{Frame-X}$ is the projections of the deflection ω^{Tri-F}_{Frame} along the X-

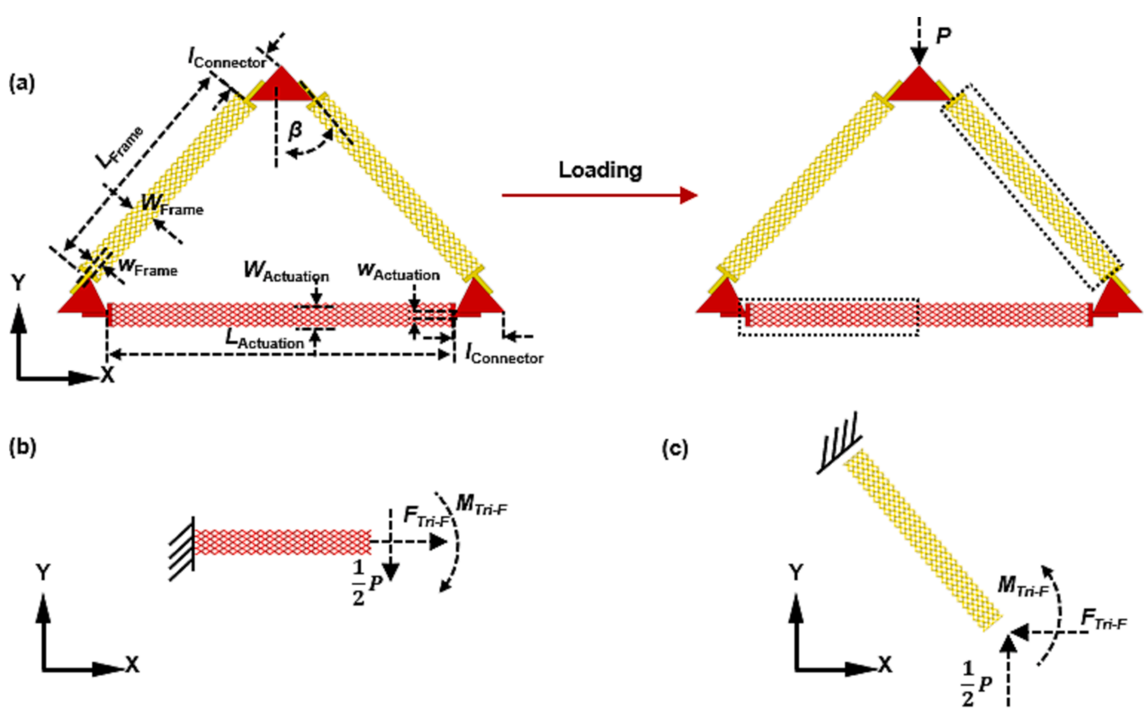

Fig. 3. Schematic illustration and theoretical analysis of the specific modulus of the triangle metamaterial. (a) FEA results of one unit cell before (left) and after (right) being subjected to a concentrated force ***P***; (b-c) Mechanical boundary conditions of the one half of horizontal beam in actuation material and an incline beam in frame material.

direction, and is $\omega_{Frame}^{Tri-F} \bullet \cos\left(\beta + \theta_{Frame}^{Tri-F}\right)$, while $\omega_{Frame-Y}^{Tri-F}$ is the corresponding projections along the Y- direction, and is $\omega_{Frame}^{Tri-F} \bullet \sin\left(\beta + \theta_{Frame}^{Tri-F}\right)$. Finally, the mechanical deformation Δ_F of the metamaterial due to the concentrated force P along the Y direction can be expressed as.

$$\Delta_F = -\omega_{Actuation}^{Tri-F} + \omega_{Frame-Y}^{Tri-F} \tag{17}$$

And the effective elastic modulus of the triangle metamaterial $E_{Triangle}$, can be expressed as.

$$E_{Triangle} = \frac{P \bullet h_{Triangle}}{\Delta_F \bullet A_{Triangle}} \tag{18}$$

Here, $A_{Triangle}$ as the effective cross-sectional area of the triangle metamaterial, and is $(L_{Actuation} + 2l_{Connector}) \bullet T_{Frame}$. Thus, the dependences of the effective elastic modulus of the triangle metamaterial on the corresponding geometric parameters are theoretically predicted. The density of the triangle metamaterial whose beams are in solid material and filled by lattice structures can be expressed as.

$$\rho_{Triangle} = \begin{cases} \dfrac{V_{Frame-S} \bullet \rho_F + V_{Actuation-S} \bullet \rho_A}{V_{Triangle}} & Solidbeams \\ \dfrac{V_{Frame-L} \bullet \rho_F + V_{Actuation-L} \bullet \rho_A}{V_{Triangle}} & Lattice-filledbeams \end{cases} \tag{19}$$

Here, ρ_F and ρ_A are densities of the frame and actuation materials. $V_{Triangle}$, the volume of the triangle metamaterial, and is $\frac{(L_{Frame}+2l_{Connector})^2 \bullet T_{Frame}}{2\cos\beta}$. $V_{Frame-S}$ and $V_{Actuation-S}$ are the actual volume of the regions in frame and actuation materials of the solid beams constructed metamaterial.

$$V_{Frame-S} = 2L_{Frame} \bullet T_{Frame} \bullet W_{Frame} + 4l_{Frame} \bullet t_{Frame} \bullet w_{Frame} \tag{20}$$

$$V_{Actuation-S} = L_{Actuation} \bullet T_{Actuation} \bullet W_{Actuation} + 2l_{Actuation} \bullet t_{Actuation} \bullet w_{Actuation} \tag{21}$$

Similarly, $V_{Frame-L}$ and $V_{Actuation-L}$ are the actual volume of the regions in frame and actuation materials of the hourglass lattice-filled beams constructed metamaterial.

$$V_{Frame-L} = 2L_{Frame} \bullet T_{Frame} \bullet \overline{\rho}_{Lattice} \bullet W_{Frame} + 4l_{Frame} \bullet t_{Frame} \bullet w_{Frame} \tag{22}$$

$$V_{Actuation-L} = L_{Actuation} \bullet T_{Actuation} \bullet \overline{\rho}_{Lattice} \bullet W_{Actuation} + 2l_{Actuation} \bullet t_{Actuation} \bullet w_{Actuation} \tag{23}$$

Here, $\overline{\rho}_{Lattice}$ represents the relative density of the hourglass lattice structure, and is $\overline{\rho}_{Lattice} = \frac{2\pi r^2}{L_{Rod}^2 \bullet \sin^2\theta \bullet \cos\theta}$. And the specific modulus of the triangle metamaterial can be given as $E_{Triangle}^* = \frac{E_{Triangle}}{\rho_{Triangle}}$.

2.4. Comparison of theoretical predictions and FEA results of the triangle metamaterial

Note that the metamaterial shown in Fig. 4 allows access to a tunable CTE range from negative to positive by tailoring the geometric parameters of each beam and the structural angle. The excellent agreements shown in Fig. 4(a) between the theoretical predictions and FEA results suggest the theoretical model as a reliable reference for the design of triangle metamaterials with a tunable thermal expansion and an ultra-low absolute value of effective CTE. Here, FEA results of the mechanical deformation and thermal response of the metamaterial were conducted through the commercial software ABAQUS, and refined meshes were adopted to ensure computational accuracy. The value of CTE ratio ($\alpha_{Triangle}/\alpha_{Frame}$) decreases nonlinearly from 1.03 to -3.06 for the metamaterials constructed by solid beams, and from 0.93 to -0.99 for the metamaterials consisting of hourglass lattice-filled beams as the angle β increases from 12.5° to 67.5°, while fixing the widths of actuation and frame beams as $10mm$. The derivation of the effective CTE of the mechanical metamaterial from positive to negative reveals the existence of a zero CTE value. Fig. 4(b)-(c) depict the contour plot of theoretical predictions of CTE ratio ($\alpha_{Triangle}/\alpha_{Frame}$) with respect to the two dominant normalized geometric parameters: the angle β and the width to length ratio (W_{Frame}/L_{Frame}) for the triangle metamaterial whose beams are in solid materials and filled by the hourglass lattice structures, respectively. Note that a set of optimized geometric parameters for the metamaterial to achieve an ultra-low CTE value (i.e., [-0.1 0.1] ppm/°C) is marked by the black line in Fig. 4(b)-(c). Here, the angle β represents the key design factor that affect the thermal expansion performances of the triangle metamaterial, and a narrow β window can be derived to render a remarkable thermal–mechanical stability. The triangle metamaterial involving hourglass lattice-filled beams has a larger β window to render a remarkable thermal–mechanical stability. As shown in Fig. 5 (a), as the angle β increases from 12.5° to 67.5°, the equivalent density $\rho_{Triangle}$ of the triangle metamaterial consisting of solid beams exhibits nonlinear fluctuations around $1.775\,g/cm^3$, with the fluctuation range of $[-2.06\% \quad 2.93\%]$. Similarly, the equivalent density $\rho_{Triangle}$ of the triangle metamaterial constructed by hourglass lattice-filled beams exhibits nonlinear fluctuations around $0.549\,g/cm^3$, with the fluctuation range of $[-3.64\% \quad 4.26\%]$. These nonlinear fluctuations are caused by the change of the actual volume (i.e., $V_{Frame-S}$, $V_{Actuation-S}$, $V_{Frame-L}$ and $V_{Actuation-L}$) of the regions in frame and actuation materials of metamaterials, and the simultaneous change of the volume (i.e., $V_{Triangle}$) of

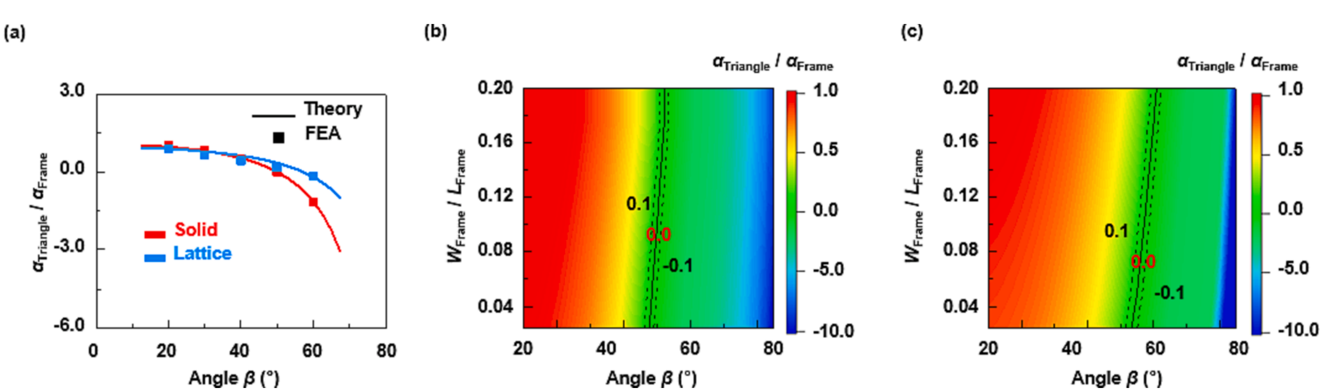

Fig. 4. Dependences of the thermal expansion of the triangle metamaterial inspired by the building block assembly method on the dominant geometric parameters. (a) Dependences of effective CTE ratio $\alpha_{Triangle}/\alpha_{Frame}$ of the triangle metamaterial on the dominant geometric parameter angle β for $W_{Frame} = 10mm$; (b) Contour plot of the theoretically predicted effective CTE ratio $\alpha_{Triangle}/\alpha_{Frame}$ in terms of angle β and W_{Frame}/L_{Frame} of the triangle metamaterial involving solid beams; (c) Contour plot of the theoretically predicted effective CTE ratio $\alpha_{Triangle}/\alpha_{Frame}$ in terms of angle β and W_{Frame}/L_{Frame} of the triangle metamaterial involving hourglass lattice-filled beams. The other parameters are $L_{Frame} = 100mm$, $W_{Actuation} = W_{Frame} = T_{Actuation} = T_{Frame} = t_{Actuation} = t_{Frame}$, $w_{Actuation} = w_{Frame} = 2mm$, $l_{Actuation} = l_{Frame} = 10mm$.

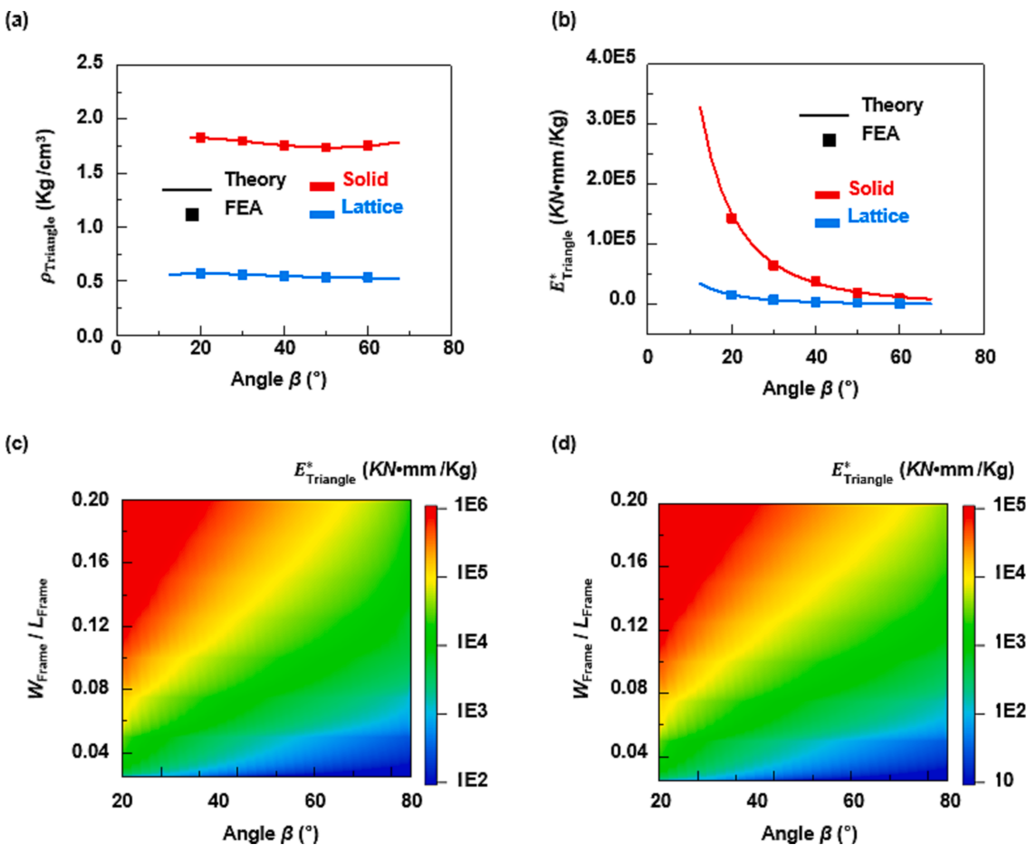

Fig. 5. Dependences of mechanical properties of the triangle metamaterial inspired by the building block assembly method on the dominant geometric parameters. (a) Dependences of the equivalent density ρ_{Triangle} of the triangle metamaterial on the dominant geometric parameter angle β for $W_{Frame} = 10mm$; (b) Dependences of the specific modulus $E^*_{Triangle}$ of the triangle metamaterial on the dominant geometric parameter angle β for $W_{Frame} = 10mm$; (c) Contour plot of the theoretically predicted specific modulus $E^*_{Triangle}$ in terms of angle β and $W_{\text{Frame}}/L_{\text{Frame}}$ of the triangle metamaterial constructed by solid beams; (d) Contour plot of the theoretically predicted specific modulus $E^*_{Triangle}$ in terms of angle β and $W_{\text{Frame}}/L_{\text{Frame}}$ of the triangle metamaterial constructed by hourglass lattice-filled beams. The other parameters are $L_{Frame} = 100mm$, $W_{Actuation} = W_{Frame} = T_{Actuation} = T_{Frame} = t_{Actuation} = t_{Frame}$, $w_{Actuation} = w_{Frame} = 2mm$, $l_{Actuation} = l_{Frame} = 10mm$.

the triangle metamaterial, with the increase of angle β. The remarkable agreements between theoretical predictions and FEA results intuitively characterize the lightweight properties of metamaterials. Additionally, the triangle metamaterial constructed by hourglass lattice-filled beams has only about 30 % of the equivalent density of the metamaterial consisting of solid beams. As evidenced by the excellent agreements between the theoretical predictions and FEA results shown in Fig. 5(b), the theoretical model can also accurately predict the specific modulus $E^*_{Triangle}$ of the triangle metamaterial. As the angle β increases in the range from 12.5° to 67.5°, $E^*_{Triangle}$ decreases from $327789.96\,kN \bullet mm/kg$ to $8676.90\,kN \bullet mm/kg$ for the solid beams constructed metamaterials, while decreases from $11481.68\,kN \bullet mm/kg$ to $317.50\,kN \bullet mm/kg$ for the metamaterial composed of hourglass lattice-filled beams, both of which reveal the remarkable capability of the triangle metamaterial in providing the high-level specific modulus. The contour plots of the specific modulus $E^*_{Triangle}$ with respect to β and W_{Frame}/L_{Frame} are shown in Fig. 5(c)-(d), revealing the dependences of the specific modulus $E^*_{Triangle}$ of the mechanical metamaterials consisting of solid and lattice-filled beams on these two dominant normalized geometric parameters, respectively. Notably, a smaller angle β but a larger width to length ratio W_{Frame}/L_{Frame} result in the higher specific modulus for the triangle mechanical metamaterial.

3. Experimental section

Fig. 6 gives a set of experiment specimen of the triangle metamaterial composed by three beams with different geometric parameters. As evidenced by Fig. 6(a), the triangle mechanical metamaterial is constructed by two solid beams in Ti, one solid beam in Al and three well-designed connectors in Ti. As shown in Fig. 6(b), the solid beam in Ti is labeled as Ti-S, and the ones in Al with different lengths are labeled from Al-S-1 to Al-S-3. And the connectors are labeled as C-T and C-B for the top and bottom connectors, respectively. Here, the solid beams of the triangle metamaterial are fabricated through the mechanical machining methods, and have the constant mechanical and thermal expansion performances (i.e., the Young's modulus: $E_{Al} = 12.16GPa$, CTE: $\alpha_{Al} = 21ppm/K$ for Al, and the Young's modulus: $E_{Ti} = 23.23GPa$, CTE: $\alpha_{Ti} = 9.58ppm/K$ for Ti). All of the connectors to join the neighboring beams are also manufactured by the mechanical machining methods, and the bolts are brought from the commercial company. Fig. 6(c) summarizes the optical images of the specimen of the mechanical metamaterials constructed by solid beams in frame and actuation materials. Similarly, Fig. 7 gives a set of experiment specimen of the triangle metamaterial consisting of lattice-filled beams with different geometric parameters. As illustrated in Fig. 7(a), the triangle metamaterial is constructed by two lattice-filled beams in Ti and one lattice-filled beam in Al. The well-designed connectors in Al are introduced to join the neighboring lattice-filled beams. As shown in Fig. 7(b), the lattice-filled beam in Ti is labeled as Ti-L, and the ones in Al with different lengths are labeled from Al-L-1 to Al-L-3. Here, the beams filled by hourglass lattice structures are fabricated with the aid of the additive manufacturing technology (i.e., selective electron beam melting (SEBM) technology). However, the modulus of the 3D printed materials are highly related to the printing angle during the additive manufacturing process as demonstrated previously [49]. Here, the printing angle θ_{Print} of the hourglass lattice is finally determined as 60°, corresponding to the Young's modulus $E = 10.43GPa$ for the printed Al and the Young's modulus $E = 19.7GPa$ for printed Ti. And the top and bottom connectors (C-T and C-B) shown in Fig. 7(c) have the same patterns with those in Fig. 6(c). Fig. 7(d) summarizes the optical images of specimen of the mechanical metamaterials constructed by lattice-filled beams but with different angles. In the experiments of this paper, the values of the angle β for these specimen in Fig. 6(d) and Fig. 7(d) are 40°, 50° and 60°. Thus, the specimen of the triangle metamaterial consisting of hourglass lattice-filled beams are constructed by the same collection of the connectors with different

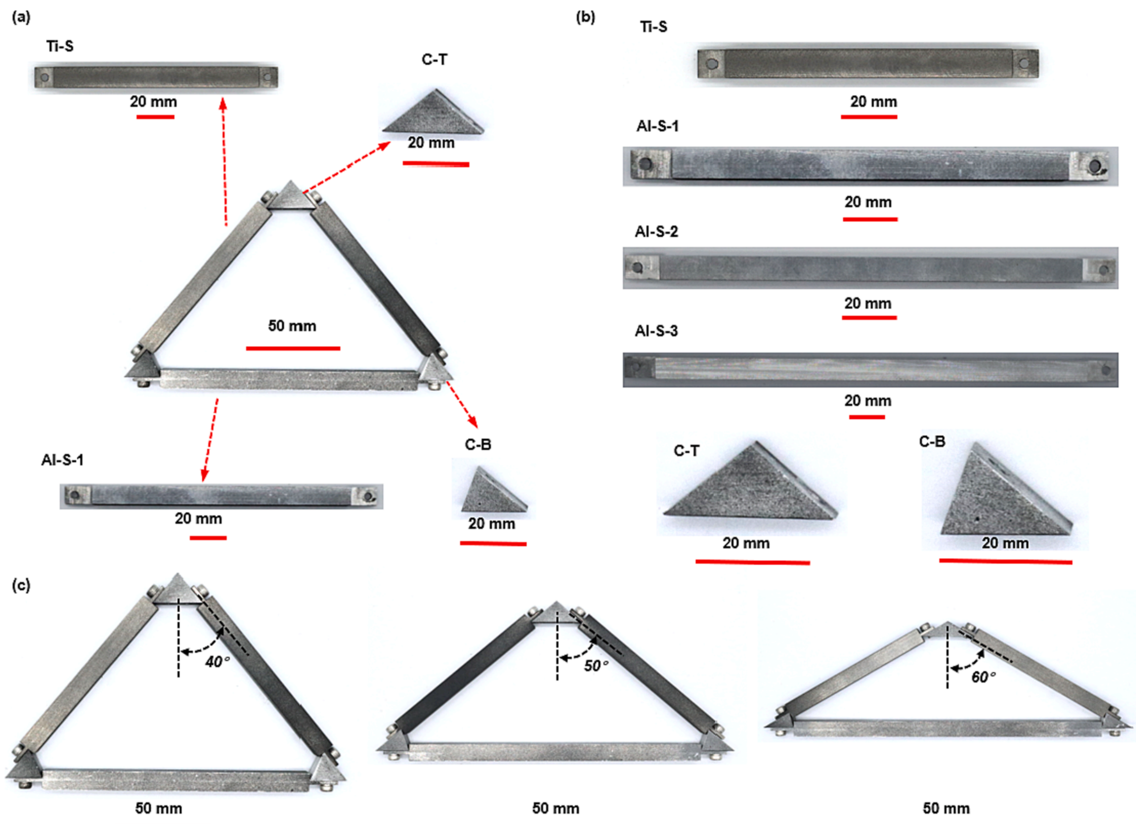

Fig. 6. Optical images of specimen in different geometric parameters of the triangle metamaterial consisting of solid beams. (a) Optical images of a triangle metamaterial inspired by the building block assembly approach, as well as its component parts; (b) Optical images of the solid beam in Ti for $L_{Frame} = 100mm$ (named as Ti-S), the solid beams in Al for $L_{Actuation} = 140.0mm, 174.5mm, 202.5mm$ (named as Al-S-1, Al-S-2, Al-S-3), and the top and bottom connectors of the triangular specimen (named as C-T and C-B); (c) Optical images of the specimen of the triangle mechanical metamaterial consisting of solid beams for the angle $\beta = 40, 50, 60^{\circ}$. The other parameters are $W_{Actuation} = W_{Frame} = T_{Actuation} = T_{Frame} = t_{Actuation} = t_{Frame} = 10mm$, $w_{Actuation} = w_{Frame} = 2mm$, $l_{Actuation} = l_{Frame} = 10mm$.

angles β from those consisting of solid beams. The length of solid and hourglass lattice-filled beams in frame material (i.e., Ti) are all fixed as $100mm$, while those of the beams in actuation material (i.e., Al) related to the configurations of the triangle metamaterial are different. Considering its weak influence on the thermal expansion of the triangle metamaterial, $W_{\mathrm{Frame}}/L_{\mathrm{Frame}}$ is fixed as 0.1. With the aid of high precision mechanical machining method and advanced additive manufacturing technology, the fabricated specimen are in good accordance with the designed configurations in the aspect of geometrical parameters.

Fig. 8(a)-(b) present the variations of thermal stain versus the temperature change for the triangle metamaterial consisting of two kinds of beams (i.e., the solid and hourglass lattice-filled beams). The thermal expansion performances of the metamaterials are measured according to the advanced platform equipped with an insulation box, a heating device and two charge-coupled-device (CCD) cameras (RTTS-100). Here, the resolution of the CCD camera can realize the measurement of deformation with an accuracy of 0.39 μm under the 100 mm sampling field of view (FOV), ensuring the measuring accuracy of thermal expansion. The environmental temperature increases from 40 °C to 200 °C, following a stepwise increment of 40 °C. And the camera takes multiple digital images at 2 s intervals to evaluate the error of thermal deformation measurement of metamaterials. The linear responses in the thermal strain of these metamaterials with respect to the temperature change are observed in the theoretical predictions, FEA results and experiments, and are beneficial for their precise control of thermal deformations in practical applications. Thus, as shown in Fig. 8(c), the experimentally measured effective CTE, derived from the slopes of experimental curves in Fig. 8(a)-(b), are $6.35\,ppm/^{\circ}\mathrm{C}$, $0.07\,ppm/^{\circ}\mathrm{C}$ and $-12.30\,ppm/^{\circ}\mathrm{C}$ for the specimen composed of solid beams, and $5.09\,ppm/^{\circ}\mathrm{C}$, $2.24\,ppm/^{\circ}\mathrm{C}$ and $-2.40\,ppm/^{\circ}\mathrm{C}$ for the ones constructed by the hourglass lattice-filled beams. Notably, $0.07\,ppm/^{\circ}\mathrm{C}$ establishes an unprecedented value observed in the experiments for offering a remarkable thermal–mechanical stability as compared to the previously reported record (e.g., $0.18\,ppm/^{\circ}\mathrm{C}$) [28]. Additionally, the specific moduli of metamaterials are experimentally measured by static loading tests on the mechanical testing platform. Fig. 8(d) evidences the dependences of the specific modulus $E^{*}_{Triangle}$ versus the angle β of the metamaterial in terms of experiments, FEA results and theoretical predictions. The experimentally measured specific moduli are $41385.55\,kN \bullet mm/kg$, $18862.85\,kN \bullet mm/kg$ and $10179.37\,kN \bullet mm/kg$ for the specimen composed of solid beams, and $3530.45\,kN \bullet mm/kg$, $1716.91\,kN \bullet mm/kg$ and $1206.34\,kN \bullet mm/kg$ for the ones constructed by the hourglass lattice-filled beams, which verifies the accuracy of the theoretical model in predicting the effective modulus of metamaterials. And the excellent agreements along theoretical predictions, FEA results, and experiment results provide the capability of the triangle metamaterial in offering a high-level thermal–mechanical stability and a high mechanical load-bearing capability.

It is worth noting that the previously researches mainly focus on how to achieve a mechanical metamaterial with a remarkable thermal–mechanical stability [14,15,23,41,49,51–53], and the one with combined attributes of an ultra-low absolute value of effective CTE and a high-level specific modulus are rarely reported. Thus, the factor, $\Phi = \phi_1 \bullet \overline{\alpha} + \phi_2 \bullet \overline{E}^{*}$, is proposed herein to quantitatively characterize the comprehensive property of our metamaterials in offering both a remarkable thermal–mechanical stability and a high mechanical load-bearing capability, as illustrated in Fig. 9. Here, $\overline{\alpha}$ and $\overline{E}^{*}$ are the two

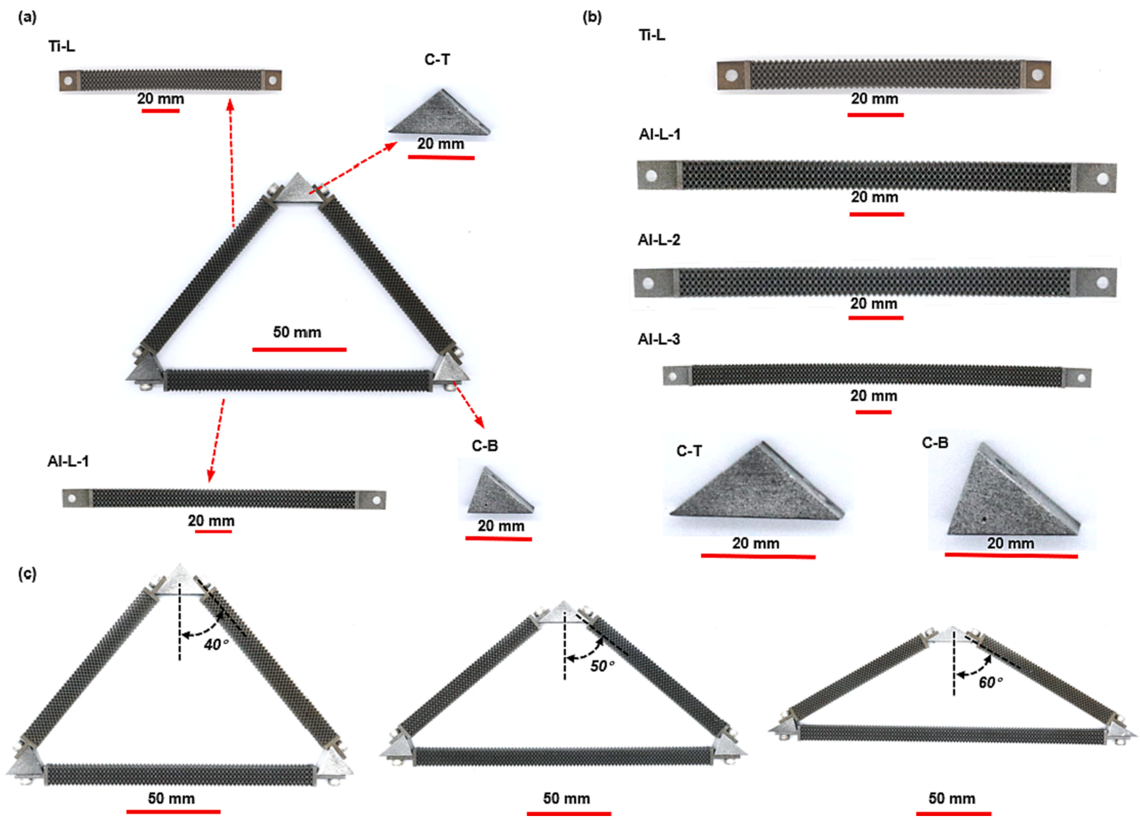

Fig. 7. Optical images of specimen in different geometric parameters of the triangle metamaterial consisting of hourglass lattice-filled beams. (a) Optical images of a triangle metamaterial inspired by the building block assembly approach, as well as its component parts; (b) Optical images of the hourglass lattice-filled beam in Ti for $L_{Frame} = 100mm$ (named as Ti-L), the hourglass lattice-filled beams in Al for $L_{Actuation} = 140.0mm, 174.5mm, 202.5mm$ (named as Al-L-1, Al-L-2, Al-L-3), and the top and bottom connectors of the triangular specimen (named as C-T and C-B); (c) Optical images of the specimen of the triangle mechanical metamaterial consisting of hourglass lattice-filled beams for the angle $\beta = 40, 50, 60^{\circ}$. The other parameters are $W_{Actuation} = W_{Frame} = T_{Actuation} = T_{Frame} = t_{Actuation} = t_{Frame} = 10mm$, $w_{Actuation} = w_{Frame} = 2mm$, $l_{Actuation} = l_{Frame} = 10mm$.

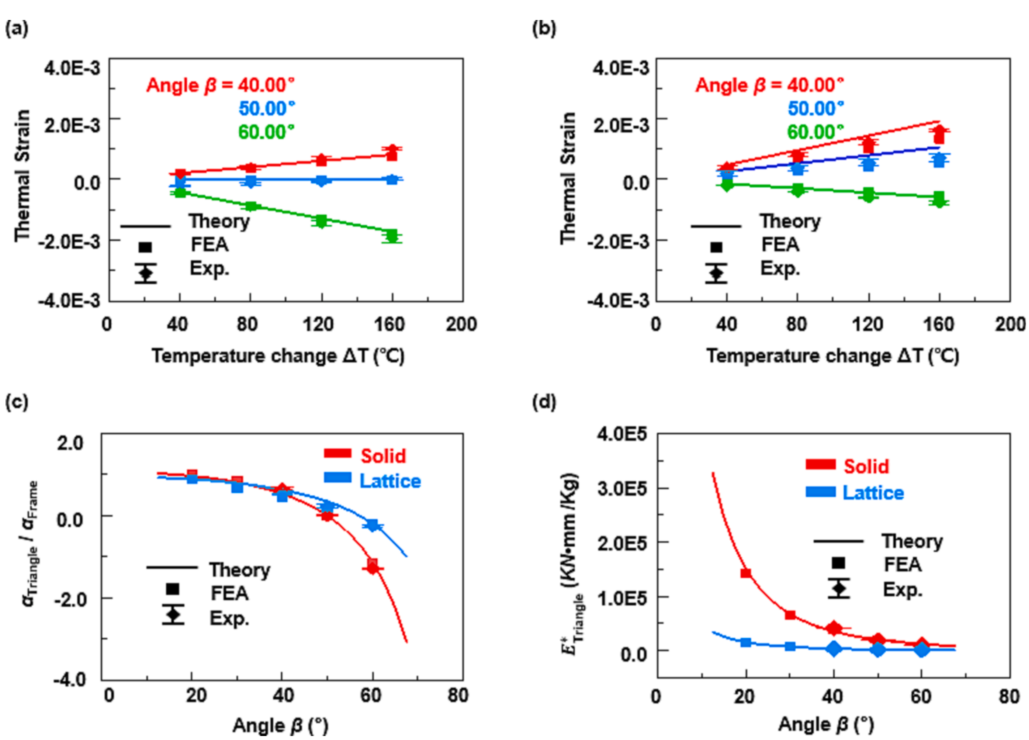

Fig. 8. Experimental demonstrations of the triangle metamaterial in different geometric parameters. (a-b) The thermal strain versus the temperature change for solid beams and hourglass lattice-filled beams constructed triangle metamaterials with different angle β; (c) The dependences of the effective CTE ratio (i.e., $\alpha_{Triangle}/\alpha_{Frame}$) versus the angle β of the two kinds of triangle metamaterials in terms of experiments, FEA results and theoretical predictions; (d) The dependences of the specific modulus $E^{*}_{Triangle}$ versus the angle β of the two kinds of triangle metamaterials in terms of experiments, FEA results and theoretical predictions.

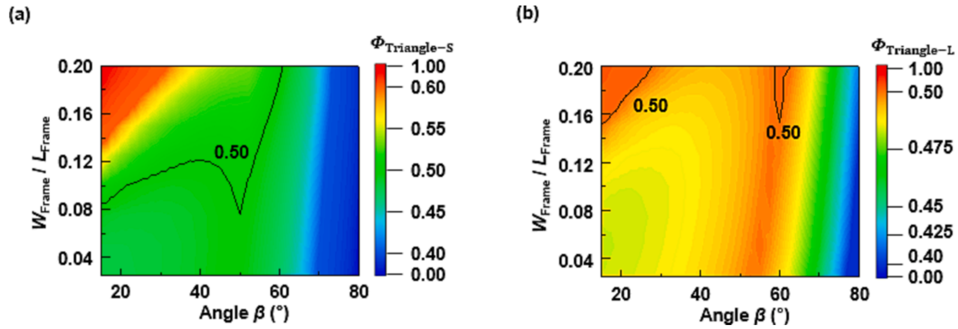

Fig. 9. The factor of mechanical properties and thermal stability $\Phi = \phi_1 \bullet \overline{\alpha} + \phi_2 \bullet \overline{E}^*$ (where $\overline{\alpha} = 1 - \frac{|\alpha_{meta}|}{|\alpha_{Max}|}$, $\overline{E}^* = \frac{E^*_{meta}}{E^*_{Max}}$, and $\phi_1 = \phi_2 = 0.5$) of the mechanical metamaterials on the dominated geometric parameters. (a-b) Contour plot of the thermal–mechanical factor in terms of the angle β and W_{Frame}/L_{Frame} for the triangle metamaterial composed of solid and hourglass lattice-filled beams.

normalized parameters, and are expressed as $1 - \frac{|\alpha_{meta}|}{|\alpha|_{Max}}$ and $\frac{E^*_{meta}}{E^*_{Max}}$, respectively. α_{meta} and E^*_{meta} are the effective CTE and specific modulus, while $|\alpha|_{Max}$ and E^*_{Max} are the maximum absolute values of the effective CTE and specific modulus of all metamaterials in this work. ϕ_1 and ϕ_2 are their corresponding weight coefficients, and are fixed as 0.5 considering the balanced requirements for the thermal–mechanical stability and the excellent mechanical property. Fig. 9 summarizes the contour plots of this factor for the triangle metamaterial composed of solid and the hourglass lattice-filled beams. Notably, a higher factor Φ is essential for metamaterials in offering both an ultra-low absolute value of effective CTE and a high-level specific modulus. Fig. 10 provides an Ashby plot of the thermal–mechanical factor Φ versus the density ρ that covers triangle metamaterials composed of solid and hourglass lattice-filled beams, metamaterials reported previously [2,13,29,49,50], and common engineering materials (i.e., metallic materials, ceramic materials, and polymer materials). Here, the results only focus on common engineering materials and experimental demonstrations of metamaterials reported previously [55]. From the Ashby plot, most of metamaterials reported previously and common engineering materials have inherent defects (i.e., high equivalent density or low thermal–mechanical factors). Comparisons between triangle metamaterials here and other materials/metamaterials effectively characterize the remarkable properties of triangle metamaterials consisting of solid beams and hourglass lattice-filled beams, such as the high-level lightweight, and excellent thermal–mechanical stability.

Fig. 11 presents the absolute value of CTEs of the triangle metamaterial in this work, as well as ones of the previously reported metamaterials whose absolute value of effective CTE range from 0 to 10.3 $ppm/°C$ [2,3,13,22,24,28–33,49,50,54]. In particular, all of CTEs summarized here are the results that are measured in the experiments, noting that many designs of metamaterials demonstrated only through FEA results and theoretical predictions are not practicable in fabrications. For the mechanical metamaterials with tunable thermal expansions, most of their absolute value of effective CTEs are out the range of 0 to 0.4$ppm/$ °C, offering a limited enhancement in achieving a remarkable thermal–mechanical stability but complex configurations that are not practicable for fabrication as compared to natural materials. Though the octet metamaterial allows access to an ultra-low absolute value of CTE (i.e., 0.18$ppm/$ °C), the narrow accessible temperature window (25 °C to 125 °C) and immutability of the specimen due to the

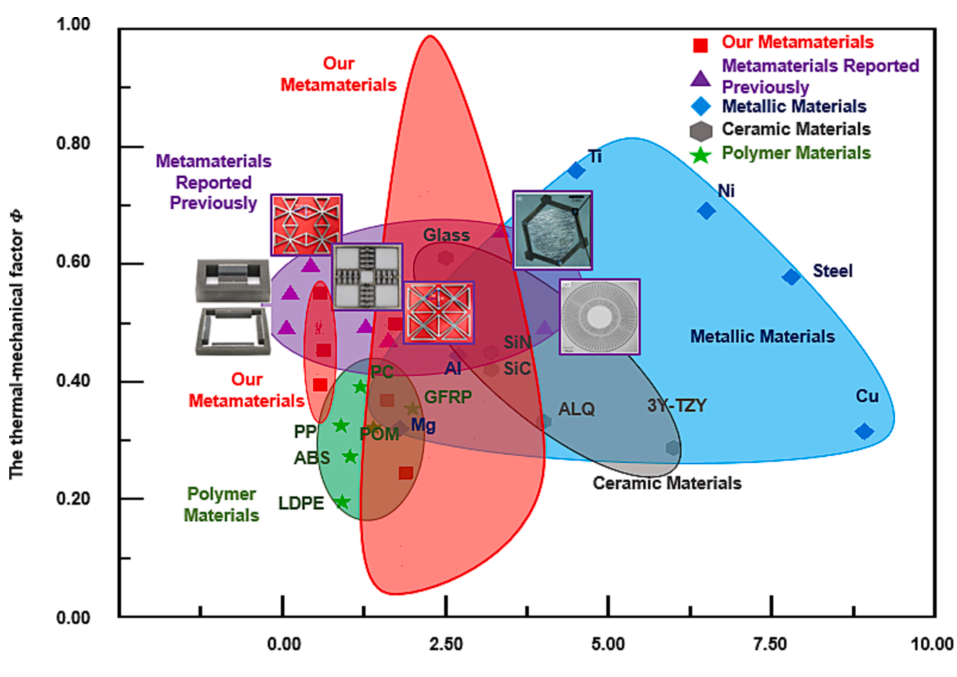

Fig. 10. Ashby plot of the density ρ and the thermal–mechanical factor Φ of our metamaterials composed of solid and the hourglass lattice-filled beams, metamaterials reported previously [2,13,29,49,50], and common engineering materials (i.e., metallic materials, ceramic materials, and polymer materials).

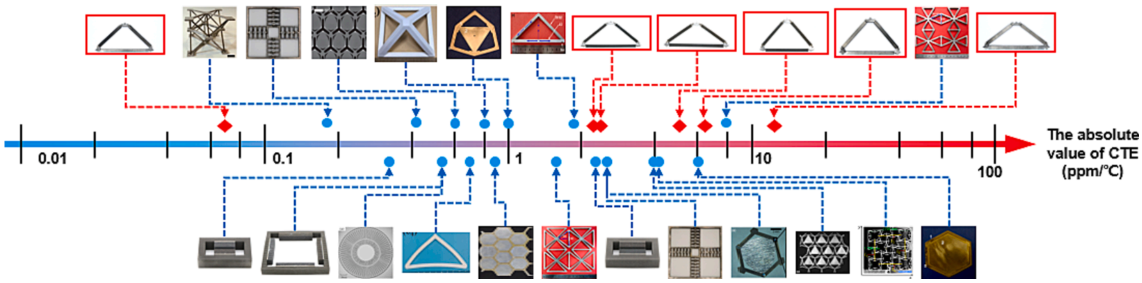

Fig. 11. The comparison of the absolute value of CTEs of the triangle metamaterial in this work with the representative mechanical metamaterials reported previously [2,3,13,22,24,28–33,49,50,54].

connection methods via superglue place certain laminations on their practicable applications in the fields that require some specific features such as low cost, recyclable and high-level thermal–mechanical stability in harsh environments [28]. The mechanical metamaterials inspired by the building block assembly method not only offer a set of unprecedented low absolute value of effective CTE ($0.07\,ppm/^{\circ}$C), but also promise the high-level mechanical load-bearing capability (i.e., $> 3 \times 10^5\ kN \bullet mm/kg$). Additionally, the unique assembly method with the well-designed connectors and bolts presented here enables the reconfigurable feature of the fabricated metamaterials to address the other potential demands.

4. Conclusions

In conclusion, the design strategies inspired by the building block assembly method for reconfigurable mechanical metamaterials with combined advantages of ultra-low CTE (i.e., <0.1 ppm/°C), lightweight and high-level specific modulus are systematically demonstrated in terms of theoretical predictions, FEA results, and experiment results. The triangle metamaterial, constructed by the specific collection of structural building blocks (i.e., solid and lattice filled beams in Ti and Al materials), show the capability of this design for achieving an unprecedented low effective CTE (i.e., 0.07 ppm/°C), high-level lightweight (i.e., 0.549 g/cm^3), a remarkable relative specific modulus (i.e., 41385.55 kN•mm/kg), and reconfigurability. Note that different from previously reported connecting methods (i.e., through dovetail connectors or super glue with strong adhesion force), the assembly method with well-designed connectors and bolts presented here not only promises the mechanical robustness of the metamaterial upon a cyclic thermal heating, but also enables the reconfigurable feature of the fabricated metamaterials to address the other potential demands. Additionally, the thermal–mechanical factor Φ is adopted for accurately characterizing compatibility of thermal and mechanical properties of the metamaterial. Finally, the comparison of the absolute value of CTE and the comprehensive thermal–mechanical property serves a quantitative comparison of the mechanical metamaterials demonstrated here to the previous reported results, indicating the capability of our designs for their practical engineering applications.

CRediT authorship contribution statement

Huabin Yu: Conceptualization, Methodology, Validation, Investigation, Writing – original draft. **Haomiao Wang:** Validation, Investigation. **Xiaogang Guo:** Supervision, Project administration, Writing – review & editing. **Bo Liang:** Methodology, Validation. **Xiaoyue Wang:** Investigation. **Hao Zhou:** Methodology, Resources. **Xiaoyu Zhang:** Methodology. **Mingji Chen:** Methodology. **Hongshuai Lei:** Investigation, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

X. G. acknowledges support from National Natural Science Foundation of China (Grant Nos. 12072030) and Young Elite Scientists Sponsorship Program by CAST. H. Z. acknowledges support from National Natural Science Foundation of China (Grant Nos. 12172041). H. L. acknowledges support from National Natural Science Foundation of China (Grant Nos. 12122202).

References

[1] Wang Q, Jackson JA, Ge Q, Hopkins JB, Spadaccini CM, Fang NX. Lightweight Mechanical Metamaterials with Tunable Negative Thermal Expansion. Phys Rev Lett 2016;117:175901.
[2] Gdoutos E, Shapiro AA, Daraio C. Thin and Thermally Stable Periodic Metastructures. Exp Mech 2013;53(9):1735–42.
[3] Rhein RK, Novak MD, Levi CG, Pollock TM. Bimetallic low thermal-expansion panels of Co-base and silicide-coated Nb-base alloys for high-temperature structural applications. Mater Sci Eng A 2011;528(12):3973–80.
[4] Pang W, Cheng Xu, Zhao H, Guo X, Ji Z, Li G, et al. Electro-mechanically controlled assembly of reconfigurable 3D mesostructures and electronic devices based on dielectric elastomer platforms. Natl Sci Rev 2020;7(2):342–54.
[5] Yan D, Chang J, Zhang H, Liu J, Song H, Xue Z, et al. Soft three-dimensional network materials with rational bio-mimetic designs. Nat Commun 2020;11(1):1180.
[6] Bai K, Cheng X, Xue Z, Song H, Sang L, Zhang F, et al. Geometrically reconfigurable 3D mesostructures and electromagnetic devices through a rational bottom-up design strategy. Sci Adv 2020;6(30):eabb7417.
[7] Shao H, Wei S, Jiang X, Holmes DP, Ghosh TK. Bioinspired Electrically Activated Soft Bistable Actuators. Adv Funct Mater 2018;28(35):1802999.
[8] Yuan C, Mu X, Dunn CK, Haidar J, Wang T, Jerry Qi H. Thermomechanically Triggered Two-Stage Pattern Switching of 2D Lattices for Adaptive Structures. Adv Funct Mater 2018;28(18):1705727.
[9] Liu J, Yan D, Zhang Y. Mechanics of unusual soft network materials with rotatable structural nodes. J Mech Phys Solids 2021;146:104210.
[10] Zhang Y, Zhang F, Yan Z, Ma Q, Li X, Huang Y, et al. Printing, folding and assembly methods for forming 3D mesostructures in advanced materials. Nat Rev Mater 2017;2(4):17019.
[11] Ma Y, Choi J, Hourlier-Fargette A, Xue Y, Chung HU, Lee JY, et al. Relation between blood pressure and pulse wave velocity for human arteries. P Natl Acad Sci USA 2018;115(44):11144–9.
[12] Zhang Y, Yan Z, Nan K, Xiao D, Liu Y, Luan H, et al. A mechanically driven form of Kirigami as a route to 3D mesostructures in micro/nanomembranes. P Natl Acad Sci USA 2015;112(38):11757–64.
[13] Xie Y, Pei X, Yu J. Double-layer sandwich annulus with ultra-low thermal expansion. Compos Struct 2018;203:709–17.
[14] Yu X, Zhou J, Liang H, Jiang Z, Wu L. Mechanical metamaterials associated with stiffness, rigidity and compressibility: A brief review. Prog Mater Sci 2018;94:114–73.
[15] Xu H, Farag A, Ma R, Pasini D. Thermally Actuated Hierarchical Lattices With Large Linear and Rotational Expansion. J Appl Mech 2019;86:111011.
[16] Chen H, Zhu F, Jang K-I, Feng X, Rogers JA, Zhang Y, et al. The equivalent medium of cellular substrate under large stretching, with applications to stretchable electronics. J Mech Phys Solids 2018;120:199–207.

[17] Zhu F, Xiao H, Li H, Huang Y, Ma Y. Irregular Hexagonal Cellular Substrate for Stretchable Electronics. J Appl Mech 2019;86(3):034501.
[18] Shi N, Song Y, Xing X, Chen J. Negative thermal expansion in framework structure materials. Coord Chem Rev 2021;449:214204.
[19] Ma Y, Zhang Y, Cai S, Han Z, Liu X, Wang F, et al. Flexible Hybrid Electronics for Digital Healthcare. Adv Mater 2020;32(15):1902062.
[20] Wei K, Peng Y, Qu Z, Zhou H, Pei Y, Fang D. Lightweight composite lattice cylindrical shells with novel character of tailorable thermal expansion. Int J Mech Sci 2018;137:77–85.
[21] Toropova MM, Steeves CA. Adaptive bimaterial lattices to mitigate thermal expansion mismatch stresses in satellite structures. Acta Astronaut 2015;113:132–41.
[22] Yamamoto N, Gdoutos E, Toda R, White V, Manohara H, Daraio C. Thin films with ultra-low thermal expansion. Adv Mater 2014;26(19):3076–80.
[23] Zhang E, Liu Y, Zhang Y. A Computational Model of Bio-Inspired Soft Network Materials for Analyzing Their Anisotropic Mechanical Properties. J Appl Mech 2018;85:071002.
[24] Steeves CA, Mercer C, Antinucci E, He MY, Evans AG. Experimental investigation of the thermal properties of tailored expansion lattices. Int J Mech Mater Des 2009;5(2):195–202.
[25] Dal H, Acikgoz K, Badienia Y. On the Performance of Isotropic Hyperelastic Constitutive Models for Rubber-Like Materials: A State of the Art Review. Appl Mech Rev 2021;73:010802.
[26] Firooz S, Steinmann P, Javili A. Homogenization of Composites with Extended General interfaces: Comprehensive Review and Unified Modeling. Appl Mech Rev 2021;73:040802.
[27] Ge Y, Zou HX, Wang S, Zhao LC, Zhang WM. Bio-Inspired Vibration Isolation: Methodology and Design. Appl Mech Rev 2021;73:4049946.
[28] Xu H, Pasini D. Structurally Efficient Three-dimensional Metamaterials with Controllable Thermal Expansion. Sci Rep 2016;6:34924.
[29] Wei K, Chen H, Pei Y, Fang D. Planar lattices with tailorable coefficient of thermal expansion and high stiffness based on dual-material triangle unit. J Mech Phys Solids 2016;86:173–91.
[30] Wei K, Peng Y, Wen W, Pei Y, Fang D. Tailorable Thermal Expansion of Lightweight and Robust Dual-Constituent Triangular Lattice Material. J Appl Mech 2017;84:101006.
[31] Li X, Gao L, Zhou W, Wang Y, Lu Y. Novel 2D metamaterials with negative Poisson's ratio and negative thermal expansion. Extreme Mech Lett 2019;30:100498.
[32] Zhang K, Wang K, Chen J, Wei K, Liang Bo, He R. Design and additive manufacturing of 3D-architected ceramic metamaterials with programmable thermal expansion. Addit Manuf 2021;47:102338.
[33] Zhang K, Wei K, Chen J, Liang Bo, Fang D, He R. Stereolithography additive manufacturing of multi-ceramic triangle structures with tunable thermal expansion. J Eur Ceram Soc 2021;41(4):2796–806.
[34] Xu H, Farag A, Pasini D. Multilevel hierarchy in bi-material lattices with high specific stiffness and unbounded thermal expansion. Acta Mater 2017;134:155–66.
[35] Xu H, Farag A, Pasini D. Routes to program thermal expansion in three-dimensional lattice metamaterials built from tetrahedral building blocks. J Mech Phys Solids 2018;117:54–87.
[36] Ha CS, Hestekin E, Li J, Plesha ME, Lakes RS. Controllable thermal expansion of large magnitude in chiral negative Poisson's ratio lattices. Phys Status Solidi B 2015;252(7):1431–4.
[37] Liu J, Gu T, Shan S, Kang SH, Weaver JC, Bertoldi K. Harnessing Buckling to Design Architected Materials that Exhibit Effective Negative Swelling. Adv Mater 2016;28(31):6619–24.
[38] Wu L, Li Bo, Zhou Ji. Isotropic Negative Thermal Expansion Metamaterials. ACS Appl Mater Interfaces 2016;8(27):17721–7.
[39] Qu J, Kadic M, Naber A, Wegener M. Micro-Structured Two-Component 3D Metamaterials with Negative Thermal-Expansion Coefficient from Positive Constituents. Sci Rep 2017;7(1):40643.
[40] Hua J, Lei H, Zhang Z, Gao C, Fang D. Multistable Cylindrical Mechanical Metastructures: Theoretical and Experimental Studies. J Appl Mech 2019;86:071007.
[41] Ni X, Guo X, Li J, Huang Y, Zhang Y, Rogers JA. 2D Mechanical Metamaterials with Widely Tunable Unusual Modes of Thermal Expansion. Adv Mater 2019;31(48):1905405.
[42] Guo X, Ni X, Li J, Zhang H, Zhang F, Yu H, et al. Designing Mechanical Metamaterials with Kirigami-Inspired, Hierarchical Constructions for Giant Positive and Negative Thermal Expansion. Adv Mater 2021;33(3):2004919.
[43] Liu X, Cao Yu, Zheng K, Zhang Y, Wang Z, Chen Y, et al. Liquid Droplet Stamp Transfer Printing. Adv Funct Mater 2021;31(52):2105407.
[44] Takezawa A, Kobashi M, Kitamura M. Porous composite with negative thermal expansion obtained by photopolymer additive manufacturing. APL Mater 2015;3(7):076103.
[45] Takezawa A, Kobashi M. Design methodology for porous composites with tunable thermal expansion produced by multi-material topology optimization and additive manufacturing. Compos B Eng 2017;131:21–9.
[46] Badrinarayanan P, Rogalski MK, Kessler MR. Carbon fiber-reinforced cyanate ester/nano-ZrW_2O_8 composites with tailored thermal expansion. ACS Appl Mater Interfaces 2012;4(2):510–7.
[47] Qiao Y, Xiao N, Song Y, Deng S, Huang R, Li L, et al. Achieving High Performances of Ultra-Low Thermal Expansion and High Thermal Conductivity in 0.5$PbTiO_3$-0.5($Bi_{0.9}La_{0.1}$)FeO_3@Cu Core-Shell Composite. ACS Appl Mater Interfaces 2020;12(51):57228–34.
[48] Ren Z, Zhao R, Chen X, Li M, Li X, Tian He, et al. Mesopores induced zero thermal expansion in single-crystal ferroelectrics. Nat Commun 2018;9(1).
[49] Yu H, Wang H, Liang B, Guo X. Metamaterials with remarkable thermal–mechanical stability and high specific modulus: Mechanical designs, theoretical predictions and experimental demonstrations. Extreme Mech Lett 2021;49:101436.
[50] Yu H, Liang B, Zhao Z, Liu P, Lei H, Song W, et al. Metamaterials with a controllable thermal-mechanical stability: Mechanical designs, theoretical predictions and experimental demonstrations. Compos Sci Technol 2021;207:108694.
[51] Surjadi JU, Gao L, Du H, Li X, Xiong X, Fang NX, et al. Mechanical Metamaterials and Their Engineering Applications. Adv Eng Mater 2019;21(3):1800864.
[52] Al-Ketan O, Abu Al-Rub RK. Multifunctional Mechanical Metamaterials Based on Triply Periodic Minimal Surface Lattices. Adv Eng Mater 2019;21(10):1900524.
[53] Bückmann T, Stenger N, Kadic M, Kaschke J, Frölich A, Kennerknecht T, et al. Tailored 3D mechanical metamaterials made by dip-in direct-laser-writing optical lithography. Adv Mater 2012;24(20):2710–4.
[54] Berger J, Mercer C, McMeeking RM, Evans AG. The Design of Bonded Bimaterial Lattices that Combine Low Thermal Expansion with High Stiffness. J Am Ceram Soc 2011;94:s42–54.
[55] Yu H, Wu W, Zhang J, Chen J, Liao H, Fang D. Drastic tailorable thermal expansion chiral planar and cylindrical shell structures explored with finite element simulation. Compos Struct 2019;210:327–38.